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Section 2
CS 3503

① $xz = (x+y)(x+y')(x'+z)$

$x(x+y) + y(x+y')\ (x'+z)$ → Distributive

$xx + y'x + yx + yy'$ → impodent + Distributive

$x + x(y'+y) + yy'$ → $y'=1$ $y=0$ used to solve

$x + x(1) + y'y$ → $x+x=1$

$x + (y'y)(x'+z)$ → $y'=1$ $y=0$ → used that to multiply

$x + (0 \cdot 1)(x'+z)$ → used distributive

$xx' + xz$ → $x=0$ $x'=1$

$(0 \cdot 1) + xz$ → solved.

$= xz$

2a $F(x,y,z) = y\,(x' + (x+y)')$

$(x+y)' = x'y'$ = Demorgan's law

$x' + x'y'$

$x'(1+y') = x'(1) = x'$ → Distributive property + identity property

$y x'$

2b $F(x,y,z) = x'yz + xz$

①

$z(x'y + x) \Rightarrow$ Distributive property

$z(x'y + x(1)) \Rightarrow$ commutative Law

$z(x'y + x(y'+y)) \Rightarrow 1 = (y'+y)$

$z(x'y + xy' + xy) \Rightarrow$ Distribute

$z((x'+x)y + xy')$ $x'+x = 1$

$z(y + xy')$

$z(y(1) + xy')$

$z(y(x'+x) + xy')$

2b $z(y(1) + xy')$

②

↳ I abandoned the distributive [illegible] took me back to the last step

$\Rightarrow$ absorption

$z(y(x+y')) = z((y+x)(y+y'))$

$z(x+y)(1) = z(x+y)$

$= zx + yz$

3a $\quad x + x'y = x + y$

$x(y+y') + x'y) \rightarrow$ using distributive/commutative property backwards

$\rightarrow$ rearrange terms

$xy + xy' + x'y = xy + x'y + xy'$

$y(x+x') + xy'$ $\quad \rightarrow$ distributive

$y(1) + xy' = (y+x)(y+y')$

$(y+x)+1 = y+x$

3b $\quad xy + x'z + yz = xy + x'z$

$xy + x'z + yz(1)$

$xy + x'z + yz(x'+x)$

$xy + x'z + yzx' + yzx$ $\quad$ } rearrange

$xy(\underbrace{1+z}_{\text{null law}}) + x'z(\underbrace{1+y}_{\text{null law}})$

$= xy + x'z$

4a $F(x,y,z) = xy' + (x'+z)'$

using demorgans law

turns into multiplication

$$(xy'(x+z))'$$

turns into addition

$$x' + (y')' + (x'z')$$

$$= x' + y + x'z'$$

4b $F(x,y,z) = (x' \quad y)(x+z)(y'+z)'$

$$= ((x'+y)(x+z)(yz'))'$$

→ simplify then take complement

$$xy' + x'z' + y' + z$$